

$$\frac{d\text{loss}}{dl_i} = \frac{d}{dl_i} \left[-\log \frac{e^{l_i \tau}}{\sum_j e^{l_j}} \right] \rightarrow -\log(g(x))$$

$$\begin{aligned} \text{where } g(x) &= \frac{e^{l_x}}{\sum_j e^{l_j}} \\ &= -\frac{1}{g(x)} \cdot \frac{e^{l_x}}{\sum_j e^{l_j}} \\ &= -\frac{\sum_j e^{l_j}}{e^{l_x}} \cdot \frac{d}{dl_i} \frac{e^{l_x}}{\sum_j e^{l_j}} \\ &= -\frac{\sum_j e^{l_j}}{e^{l_x}} \left[0 \cdot \frac{1}{\sum_j e^{l_j}} + -\frac{e^{l_x} e^{l_j}}{(\sum_j e^{l_j})^2} \right] \\ &= \frac{e^{l_j}}{\sum_j e^{l_j}} = \rho_i \end{aligned}$$

$$\begin{aligned} \mu &= \frac{1}{m} \sum x_i \\ \sigma^2 &= \frac{1}{m\alpha} \sum_i (x_i - \mu\beta)^2 \\ \hat{x}_i &= \frac{x_i - \mu\beta}{\sqrt{\sigma_\beta^2 + \epsilon}} \\ \gamma_i &= \tau \hat{x}_i + \beta \end{aligned}$$

1)

$$\frac{dL}{d\hat{x}_i} = \frac{dL}{dy_i} \gamma$$

2)

$$\begin{aligned} \frac{dL}{d\sigma^2} &= \frac{2(x_i - \mu\beta) \frac{dL}{d\gamma_i} \gamma_i}{\sigma} = \sum_i \frac{dL}{dx_i} \frac{d\hat{x}_i}{d\sigma^2} \\ &= \gamma \sum_i [(x_i - \mu\beta)(\sqrt{\sigma_\beta^2 + \epsilon})] \frac{dL}{d\gamma_i} \\ &= \gamma \sum_i [(x_i - \mu\beta) - \frac{1}{2}(\sigma_i^2 + \epsilon)^{3/2}] \frac{dL}{d\gamma_i} \end{aligned}$$

3)

$$\begin{aligned}
\frac{\partial L}{\partial \mu} &= \sum_i \frac{\partial L}{\partial x_i} \frac{\partial L}{\partial \mu} + \frac{\partial L}{\partial \sigma^2} \frac{\partial \sigma^2}{\partial \mu} \\
&= \gamma \sum_i \frac{\partial L}{\partial \mu} + \frac{\partial L}{\partial \sigma^2} \frac{\partial \sigma^2}{\partial \mu} \\
&= \gamma \sum_i (\sigma^2 + \epsilon)^{-1/2} + \frac{\partial L}{\partial \sigma^2} \frac{\partial}{\partial \mu} \left(\sum_{i=1}^m (x_i - \mu)^2 \right) \\
\frac{\partial L}{\partial \mu} &= \gamma \sum_i (\sigma^2 + \epsilon)^{-1/2} \frac{\partial L}{\partial y_i}
\end{aligned}$$

$$\begin{aligned}
&\frac{2}{m-1} \sum_{i=1}^m (x_i - \mu) \\
&\frac{-2}{m-1} \sum_{i=1}^m (x_i - \mu) \\
&\frac{-2}{m-1} (\sum x_i - \sum \mu) \\
&\frac{-2}{m-1} (m\mu - m\mu) \\
&= 0
\end{aligned}$$

4)

$$\begin{aligned}
\frac{\partial L}{\partial x_i} &= \frac{\gamma}{m} \frac{\partial L}{\partial \hat{x}_i} \frac{\partial \hat{x}_i}{\partial x_i} + \frac{\partial L}{\partial \sigma^2} \frac{\partial \sigma^2}{\partial x_i} + \frac{\partial L}{\partial \mu} + \frac{\partial \mu}{\partial x_i} \\
&= \gamma \frac{\gamma}{m} (\sigma^2 + \epsilon)^{-1/2} + (x_i - \mu)(\sigma^2 + \epsilon)^{-3/2} +
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \hat{x}_i}{\partial x_i} &= (\sigma^2 + \epsilon)^{-1/2} \\
\frac{\partial \sigma^2}{\partial x_i} &= \frac{2}{m-1} (x_i - \mu) \\
\frac{\partial \mu}{\partial x_i} &= \frac{1}{m}
\end{aligned}$$

$$\begin{aligned}
&= \frac{\partial L}{\partial y_i} \gamma (\sigma^2 + \epsilon)^{-1/2} + \gamma \sum_j \left[(x_i - \mu)(\sigma^2 + \epsilon)^{-3/2} \right] \\
&= \left(\frac{\sum_{i=1}^m (x_i - \mu)}{m} \right) + \gamma \sum_j (\sigma^2 + \epsilon)^{-1/2} \frac{\partial L}{\partial y_i} \cdot \frac{1}{m} \\
&= \frac{\gamma (\sigma^2 + \epsilon)^{-1/2}}{m} \left[m \frac{\partial L}{\partial y_i} - \sum_j \frac{\partial L}{\partial y_j} - \frac{m}{m-1} \hat{x}_i \sum_j \frac{\partial L}{\partial y_i} \hat{x}_j \right]
\end{aligned}$$